

1. Administrative & Study Identification

- **Full Study Title:** An Extension Study of Bomedemstat (IMG-7289/MK-3543) in Participants With Myeloproliferative Neoplasms (IMG-7289-CTP-202/MK-3543-005)
 - **Short Title/Acronym:** IMG-7289-CTP-202 / MK-3543-005
 - **Protocol Number:** IMG-7289-CTP-202 / MK-3543-005
 - **NCT Number:** NCT05223920
 - **Sponsor Name:** Imago BioSciences, Inc., a subsidiary of Merck & Co., Inc.
 - **Investigational Product:** Bomedemstat (IMG-7289 / MK-3543)
 - **Phase of Development:** Phase 2
 - **Medical Monitor:** Merck & Co. Clinical Operations Lead
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2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To assess the long-term safety and efficacy of bomedemstat administered orally once daily in participants with a Myeloproliferative Neoplasm (MPN) who participated in a prior bomedemstat feeder study. * **Secondary Objectives:** To evaluate continued clinical benefit, durability of response, and overall survival in participants transitioning from primary treatment periods.

2.2 Background & Rationale To address the need for long-term safety and efficacy data collection for patients with MPN who are deriving clinical benefit from bomedemstat. Participants completing initial treatment periods in feeder studies such as IMG-7289-CTP-102 (NCT03136185) and IMG-7289-CTP-201 (NCT04254978) require an extension protocol to continue receiving the therapeutic benefits of the investigational product.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Single-arm (Open-label extension) * **Blinding:** Open-label * **Randomization:** N/A

3.2 Planned Enrollment & Duration * **Target \$N\$:** 81 participants * **Number of Sites:** 19 locations (Multi-center) * **Estimated Study Start:** 15-Dec-2021 * **Estimated Primary Completion:** 22-Aug-2024

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Has completed at least one Treatment Period (TP) in a prior bomedemstat Myeloproliferative Neoplasms (MPN) protocol (such as, but not limited to, IMG-7289-CTP-102/MK-3543-002 [NCT03136185] and IMG-7289-CTP-201/MK-3543-003 [NCT04254978]). 2. In the estimation of the Investigator, the risk-benefit favors continued dosing with bomedemstat.

4.2 Exclusion Criteria 1. Ongoing participation in another investigational study (except observational studies). 2. A history of non-compliance in a prior bomedemstat study (excluding dose suspensions that were medically warranted). 3. Current use of a prohibited medication (e.g., romiplostim). 4. Medical, psychiatric, cognitive, or other conditions that, in the Investigator's opinion, compromise the patient's safety, ability to give informed consent, or comply with the trial protocol. 5. Is pregnant or breastfeeding or plan to become pregnant. Women of childbearing potential (WOCBP) and fertile men unwilling to agree to use an approved method of contraception from time of enrollment until 14 days after last bomedemstat dose.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency:** Bomedemstat administered once daily. * **Route of Administration:** Oral capsule. * **Duration of Treatment:** 169 days, with additional treatment continuing in patients deriving clinical benefit.

5.2 Concomitant & Prohibited Therapy * **Permitted:** Standard concomitant medications for MPN complications not strictly prohibited by the protocol. * **Prohibited:** Romiplostim and other specified investigational agents.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -14 to 0	Rollover Consent from feeder study, baseline laboratory and clinical assessments, review of prior compliance
Baseline	Day 0	First extension dose, verification of continued clinical benefit from prior study
Interim	Up to 169 days	Safety labs, efficacy assessments, TEAEs monitoring, ongoing drug dispensing
End of Study	Post-169 days	Final vital signs, exit interview, transition to further extension study (e.g., MK-3543-017) if eligible

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** Long-term safety and tolerability profile evaluated by the incidence of Treatment-

Emergent Adverse Events (TEAEs) throughout the extension phase. * **Measurement Method:** Clinical assessments, laboratory evaluations, and investigator reporting.

7.2 Secondary & Exploratory Endpoints * Durability of response (e.g., spleen volume reduction, hematologic remission, or symptom score improvements depending on MPN subtype). * Overall survival (OS) and duration of clinical benefit in prolonged bomedemstat exposure.

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Standard collection via investigator assessment during scheduled visits.
 - **Serious Adverse Events (SAEs):** Safety monitoring reported within 24 hours of site awareness.
 - **Data Monitoring Committee (DMC):** Internal/External safety review to monitor the long-term risk-benefit profile.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Safety Set (all rollover participants receiving at least one dose of extension study drug).
 - **Sample Size Justification:** $N=81$, based on the number of eligible rollover participants from the designated feeder studies.
 - **Handling of Missing Data:** Last Observation Carried Forward (LOCF) for efficacy endpoints; descriptive statistics and frequency distributions for safety data.
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
 - **Monitoring Plan:** Routine onsite and remote monitoring visits to ensure data integrity and protocol adherence during the extension phase.
 - **Audit Trail:** eCRF completion, tracking of participant continuation/transition from feeder protocols, and strict monitoring of drug dispensing.
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11. Study Identifiers & Governance

- **Primary ID:** IMG-7289-CTP-202 / MK-3543-005
- **Registry Number:** NCT05223920
- **Sponsor/Lead Organization:** Imago BioSciences, Inc., a subsidiary of Merck & Co., Inc.

- **Study Title:** An Extension Study of Bomedemstat (IMG-7289/MK-3543) in Participants With Myeloproliferative Neoplasms (IMG-7289-CTP-202/MK-3543-005)
 - **Official Regulatory Title:** A Multi-Center, Open Label, Extension Study Evaluating the Safety and Efficacy of Bomedemstat for the Treatment of Patients with Myeloproliferative Neoplasms (MPNs) Enrolled in a Prior Bomedemstat Clinical Study
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12. Clinical Framework (Myeloproliferative Neoplasms Specific)

- **Primary Condition:** Myeloproliferative Neoplasms (MPN), including Primary Myelofibrosis and Essential Thrombocythemia
 - **Diagnosis Threshold:** Confirmed MPN and successful completion of a prior bomedemstat TP
 - **Medical Methodology:** LSD1 (Lysine-specific demethylase 1) inhibitor therapy via bomedemstat
 - **Optimization Target:** Maintenance of hematologic remission, clinical benefit, and long-term symptom control
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13. Study Procedures & Methodology

- **Management Pathway:** Long-term extension maintenance therapy
 - **Education Protocol (HCP):** Investigator training on long-term safety monitoring, AE management, and dose adjustment
 - **Education Protocol (Patient):** Counseling on long-term oral therapy compliance and required contraception use
 - **Quality Audit Mechanism:** Investigator risk-benefit estimation tracking and continuous protocol compliance reviews
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14. Observation & Follow-Up Schedule

- **Total Duration:** 169 days + additional treatment for those deriving benefit
 - **Onsite Visit Cadence:** Periodically as determined by the extension protocol
 - **Remote Monitoring Frequency:** Ongoing safety check-ins and AE tracking
 - **Checklist Review Interval:** Regular intervals to assess the continued risk-benefit profile
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15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	____	[DD-MMM-YYYY]	[]			